Chapter 7. The Sieve

You Shall Not Overcount

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7.1 Enumerating the elements of intersecting sets

Motivation. Adding the sizes of overlapping sets counts common elements several times. Alternately subtracting and adding back the sizes of intersections corrects every overcount exactly; the reason is that $\sum_j (-1)^j \binom{s}{j} = 0$.

Example 7.1

14 students play soccer, 17 basketball, and 4 both. Then $14 + 17 - 4 = 27$ play at least one of the two.

Example 7.2

Add 18 hockey players; 4 play soccer and basketball, 3 soccer and hockey, 5 basketball and hockey, and 1 all three. Then $14 + 17 + 18 - 4 - 3 - 5 + 1 = 38$ students play at least one game. The student playing all three is counted three times, subtracted three times, and must be added back once.

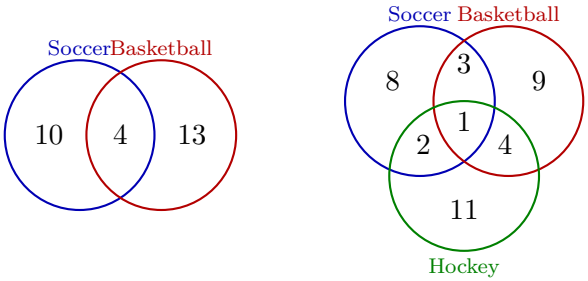


Figure. *Examples 7.1 and 7.2: the counts in the regions of the Venn diagrams sum to* 27 *and* 38.

Theorem 7.3 (Sieve formula, inclusion–exclusion)

For finite sets $A_1, \dots, A_n$,

$$|A_1 \cup \cdots \cup A_n| = \sum_{j=1}^{n} (-1)^{j-1} \sum_{\{i_1,\dots,i_j\}} |A_{i_1} \cap \cdots \cap A_{i_j}|, \tag{7.1}$$

where $\{i_1, \dots, i_j\}$ runs over the j-element subsets of $[n]$.

Proof. An element outside the union is counted by no term. Let x be in the union, $S := \{i \mid x \in A_i\}$ and $s := |S| \geq 1$. An intersection $A_{i_1} \cap \cdots \cap A_{i_j}$ contains x iff $\{i_1, \dots, i_j\} \subseteq S$, so x is counted

$$s - \binom{s}{2} + \binom{s}{3} - \cdots + (-1)^{s-1} \binom{s}{s} = 1 - (1-1)^s = 1 \tag{7.2}$$

times, by the binomial theorem. ∎

7.2 Applications of the sieve formula

Motivation. Counting objects with *none* of several bad properties is the typical use of the sieve: subtract the objects with at least one bad property, which inclusion–exclusion counts through intersections that are easy to count. Derangements and surjections are the classic cases; the second gives the promised formula for $S(n, k)$.

Example 7.4 (Derangements)

n guests pick up their hats at random. In how many ways does nobody get their own hat, i.e. how many permutations of $[n]$ have no fixed point (**derangements**)?
Solution. Let A_i be the set of permutations fixing i. An intersection of j of the A_i fixes j given points and permutes the rest freely, so it has $(n-j)!$ elements, and there are $\binom{n}{j}$ such intersections. By (7.1),

$$|A_1 \cup \cdots \cup A_n| = \sum_{j=1}^{n} (-1)^{j-1} \binom{n}{j} (n-j)! = \sum_{j=1}^{n} (-1)^{j-1} \frac{n!}{j!},$$

so the number of derangements is

$$D(n) = n! - |A_1 \cup \cdots \cup A_n| = \sum_{i=0}^{n} (-1)^i \frac{n!}{i!}. \tag{7.3}$$

Hence $\frac{D(n)}{n!} = \sum_{i=0}^{n} \frac{(-1)^i}{i!} \to e^{-1}$: roughly 37% of all permutations are derangements. [Supplement] In fact $D(n)$ is the integer nearest to $n!/e$ for $n \geq 1$: the alternating tail $\left|\frac{n!}{e} - D(n)\right| = n! \left|\sum_{i>n} \frac{(-1)^i}{i!}\right|$ is at most its first term $\frac{n!}{(n+1)!} = \frac{1}{n+1} \leq \frac{1}{2}$, with strict inequality since the terms decrease strictly.

Theorem 7.5

For positive integers n, k:

$$S(n,k) = \frac{1}{k!} \sum_{i=0}^{k} (-1)^i \binom{k}{i} (k-i)^n.$$

Proof. By Corollary 5.9, $k!\,S(n,k)$ is the number of surjections $[n] \to [k]$. Of the k^n functions, let A_i be those missing i in their image; an intersection of j such sets consists of functions into a $(k-j)$-element set, so it has $(k-j)^n$ elements. By the sieve, the number of non-surjective functions is $\sum_{j=1}^{k} (-1)^{j-1} \binom{k}{j} (k-j)^n$. Subtract this from k^n. ∎

Theorem 7.6 (Möbius inversion on subsets)

Let f, g be real functions on the subsets of $[n]$ with $g(S) = \sum_{T \subseteq S} f(T)$. Then

$$f(S) = \sum_{T \subseteq S} g(T) (-1)^{|S-T|}. \tag{7.4}$$

Proof. Expanding each $g(T)$, the right side of (7.4) is $\sum_{U \subseteq S} f(U) \sum_{U \subseteq T \subseteq S} (-1)^{|S-T|}$. A set T with $U \subseteq T \subseteq S$ is determined by $S \setminus T \subseteq S \setminus U$, and there are $\binom{|S-U|}{i}$ choices with $|S-T| = i$. So the inner sum is $\sum_i (-1)^i \binom{|S-U|}{i} = (1-1)^{|S-U|}$, which is 0 unless $U = S$, when it is 1. Hence the right side is $f(S)$. ∎

(This is the prototype of Möbius inversion on partially ordered sets.)